

Mohnish Nanthakumar

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EDUCATION:

University of Wisconsin-Madison
B.S. Computer Engineering

Madison, WI
Expected Graduation: Dec 2027

WORK EXPERIENCE:

Incoming Embedded Software Engineer Intern | Plexus Corp

Aug 2026 – Dec 2026

- Will design **real-time software** for **highly-regulated industries** (e.g. aerospace, health).
- Will implement software on microcontrollers using **Modern C++** and **Python**.
- Tasked with developing robust unit tests (**GoogleTest**) to validate real-time operating systems (RTOS).

Incoming Embedded Systems Applications Engineering Intern | DigiKey

May 2026 – Aug 2026

- Will architect, design and test a self-directed embedded systems project utilizing warehouse components including **FPGAs** and **Embedded Linux SBCs**.

TEAM PROJECTS:

Avionics Engineer | Wisconsin Space Program (WiSP)

Sep 2024 – Dec 2025

- Programmed the **Flight Computer** and **Ground Station** on an ESP32 system in C++ to transmit **mission-critical telemetry data** (e.g. altitude, acceleration) via LoRa RF.
 - Developed **unit test scripts** for every major function (e.g., LoRa initialization, Altimeter status) to validate **system stability** prior to deployment.
 - **Documented** system architecture, hardware connections, and software data flows, to ensure **strict maintainability**.
- Designed the **Engine Control Unit** (ECU) of a bipropellant liquid rocket (IPA + O₂) – Arduino sending engine data (e.g. temperature, pressure) to a ground station.
 - Designed and Documented the **Printed Circuit Board** (PCB) for the ECU.

Software Engineer | FRC Team 3130 (High School Robotics Team)

Sep 2022 – May 2024

- Developed Java-based **control software** and **embedded software** for 120+ lb, 12V competition robots.
- **Programmed, tested and tuned** new features within a **3-month timeframe**.
 - Used **PID motor control algorithms** to ensure precise, real-time robotic movements.
- Conducted lessons on Git version control, GUI-based debugging, and object-oriented programming in Java.

AWARDS:

Best Community Hack (Out of 114 Teams) | **MadHacks** (Sponsors: Google, Capital One)

- Developed a logistics web application that streamlines cross-country moving by scraping U-Haul data and estimating multi-modal travel costs
- Implemented an **A* shortest path algorithm** to process US highway data and optimize routing for trips.
- Analyzed cost datasets across airlines and gas stations to **dynamically calculate** average prices per mile.

TECHNICAL SKILLS:

Software: C, C++, Unix, Linux, Bash, Python, Make, Java, git, gdb

Embedded: FreeRTOS, Bare-Metal Programming, STM32, ESP32, Raspberry Pi, Arduino, UART, SPI, I2C

Electrical: PCB Design, Multimeter, Oscilloscope, Soldering, Wiring

Miscellaneous: MATLAB, Onshape, Fusion 360, 3D-Printing, MS Office, Slack